Route Optimization by using Dijkstra's Algorithm for the Waste Management System

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ABSTRACT

Unplanned waste collection causes environmental pollution, cost increment and large consumption of fuel. Optimized route planning is one of the most important factors in the smart waste management system (WMS). In this work, an optimal route planning model and algorithm based on Dijkstra's algorithm are proposed. For the link-cost calculation, real-life parameters such as the bins' status (fill-levels), road congestion status, distance are considered. Data analysis is done based on the practical scenario. The proposed model is found more cost-effective and Eco-friendly than the conventional model.

CCS Concepts

• **Networks→Network management.**

Keywords

Smart waste bin; smart waste management system, route optimization; Dijkstra's algorithm; smart & green city.

1. INTRODUCTION

In Malaysia, around 38000 tons of waste is generated daily [1] and the number is increasing as the population growth is increasing. One the other hand, in Malaysia, CO_2 emission was recorded in 2018 as 251.52M mt, which had a 7.21% of growth rate [2]. These two statistics are very alarming for our eco-system. Unplanned and inappropriate waste management leads to an unhealthy environment and creates a threat to our lives. A smart and planned waste management system (WMS) is highly needed to achieve green and smart cities.

Traditionally, in Malaysia, waste bins are emptied at certain intervals by the responsible stakeholder and sometimes the waste collection processes are not well scheduled. This causes some drawbacks where some waste bins fill up much faster than the rate of emptying and they are full before the next collection. This problem has consequences where the waste is overflowing and poses an unpleasant view and odor, and hygiene as well as health risks. The impacts of the waste in our lives were described in detail in [3]. There are some special periods such as festivals when the waste bins fill up very quickly and some are not at other places. This situation is a challenge for the stakeholder to maintain a clean city [4].

There are several elements can be optimized in WMS such as route optimization (to find the best route), labor cost optimization (how many labors will be needed for the waste collection), optimized schedule (the optimum timing for the waste collection), waste collecting truck requirement optimization (how many and which size of trucks are needed and so on), etc. In this work, route optimization and truck requirement optimization has been focused. In conventional WMS, waste has been collected on a daily basis. This is not an efficient system. On the other hand, in some smart WMS, waste has been collected when there is an overflow of the bins. That means waste is collected on an immediate or emergency basis. This is also not an efficient system [5]. In this paper, we have combined both the issues. In this work, waste collection will be done based on the bin's status as well as the distance, road condition (congestion), truck's capacity and so on.

Dijkstra's algorithm [6] is a very popular algorithm in computer science used to solve the single-source shortest paths for a given graph with nonnegative edge weights. In our proposed model and the algorithm, this algorithm has been applied. We have formulated the route from the garage, dumping station and the bins as the computer networking node problem. The link-cost of the routes has been calculated based on the bins' status (fill-levels), distance and road congestion. The garbage truck will follow the shortest path for the waste collection.

The contribution and the advantages of the proposed works compared to other related works:

- Various real-time and practical parameters such as distance between the garage and the bins, the status of the bins, the congestion of the road, etc. have been considered for the link cost calculation.
- The garage, dumping station and the bins are considered as the network nodes and hence simple Dijkstra's algorithm has been applied for the optimized route calculation.
- Some real values have been used for the data analysis such as the cost of the fuel, fuel economy of the truck, CO_2 emission and so on.

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- It can be applied with any other WMS network, only need to change the value of the parameters, it does not need any complex tool or algorithm.

The organization of the paper is as follows: section 1 introduces the idea of the route optimization and other optimization requirements in WMS. Section 2 discusses some related works while section 3 discussed the main idea of the work, link-cost calculation method, the proposed model and algorithm. Section 4 discusses the implementation of the algorithm, results, and analysis of the performances.

2. RELATED WORKS

Several works have been done in the route optimization for WMS. In [7], the authors used Life Cycle Assessment (LCA) technique and heuristic approach to solve the vehicle routing problem. They considered the available working time of each collection truck to assign them collection routes and minimize the number of compactor trucks.

Xue and Cao in [8] proposed an ant colony optimization (ACO)-based multi-objective routing model coupled with Dijkstra's algorithm to find the best route from these waste-generating points to the dumping station considering travel time, accident probability (black spots), and population exposure.

In [9], the authors proposed a modified Backtracking Search Algorithm (BSA) to solve vehicle routing problem models with the smart bin concept to find the best-optimized waste collection route solutions.

The authors in [10] described a case study on a real-life waste collection problem in Eastern Finland. They presented a conceptual model based on Dijkstra's algorithm waste collection route optimization. Other related works in this domain can be found in [11]–[13].

These works considered either only road condition or the distance, or the bins' fill level, but not all the aspects. They even did not consider the fuel cost and CO_2 emission from a practical point of view.

3. PROPOSED MODEL & ALGORITHM

As a case study, in Figure 1, a road map of the bins, the garage, and the dumping station has been considered. For simplicity, the garage, the dumping station, and the bins are assumed as the nodes in the graph theory. The garage has been considered as node 0 and the dumping station as node 9, and the bins' nodes are as numbered. For the cost calculation, three parameters have been considered. They are i) the distance between the nodes, ii) the status of the bin (what % it is full), and iii) the congestion of that road. Following equation has been proposed for the link-cost calculation:

$$C = \frac{\alpha D + \beta R_C}{\gamma B_S} \quad (1)$$

Here, D is the distance between the nodes, R_C is the numerical value of the road congestion, B_S is the status of the bin (% occupied or full). α, β, and γ are the tuning factors for the parameters. The sum of these values must be 1. For example, here we have chosen, $\alpha = 0.5$, $\beta = 0.3$, and $\gamma = 0.2$. That means, we have given the highest priority to the bin's status, then the distance of the road and the least for the congestion. These values can be tuned according to the requirement for the link-cost calculation. The hypothesis for selecting the tuning factors is to ensure the priority of the parameters. All the parameters' priority is not the same. For example, the status of the bins has higher priority over the congestion of the road, but it is also true that we cannot avoid the congestion in our calculation. Therefore, these tuning factors are very much important to be considered. If we don't use these factors, all the parameters' importance would be the same, and that is not feasible.

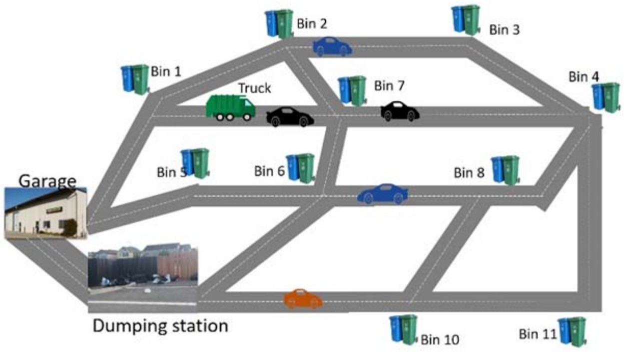


Figure 1. Road map of the bins, garage and dumping station.

Distance is measured in meter and can be measured through the GIS software from the map, IoT (Internet of Thing) based sensors will be used in the bin to measure the status of the bins [14] and the congestion can be measured based on the average speed of the vehicle and the vehicle density. These two values can be measured through the speedometer and the counter sensor by using a laser or ultrasonic sensors. We have calculated the congestion values by using the following concepts:

Table 1. Congestion value calculation

The average speed of the vehicles	Congestion status	Numerical value
0~10 km/h	High congestion	70
11~15 km/h	Medium congestion	50
16~ km/h	Low congestion	30

After calculating the cost, it would be normalized by multiplying with 3 and ceiling the value with the next integer value. These normalized values will be used as the link-cost of the paths for Dijkstra's Algorithm.

After the first step of optimal route selection, the garbage truck will follow the route and collect the bins' waste. After that, the collection bins will be omitted from the graphs. Then again the link-cost will be calculated and another round of Dijkstra's algorithm will be applied. The process will be repeated until all the nodes are visited. The complete algorithm has been given in Algorithm 1:

Algorithm 1

```
for (until all the nodes visited){
    link-cost calculation based on eq. (1)
    Run the Dijkstra's algorithm (Algorithm 2)
    Select the best route and count the nodes selected
    if (number of nodes<=3)
        Use small truck
    else
        Use big truck
    Remove the visited nodes
```

10. *Go to step 2 }*

Algorithm 2 (Dijkstra's algorithm)

```
1.   function dijkstra(G, S) {
2.     for each vertex V in G
3.        distance[V] <- infinite
4.        previous[V] <- NULL
5.        if V != S, add V to Priority Queue Q
6.     distance[S] <- 0
7.   while Q IS NOT EMPTY
8.     U <- Extract MIN from Q
9.      for each unvisited neighbor V of U
10.       tempDistance<-distance[U]+ edge_weight(U, V)
11.       if tempDistance < distance[V]
             distance[V] <- tempDistance
             previous[V] <- U
12.  return distance[], previous[]; }
```

To evaluate the proposed model and the algorithm, we have considered three cases; i) only a big truck is used to collect the waste, ii) only small truck is used, and iii) both small and big truck are used based on the proposed model and algorithm. We have assumed that small trucks can accommodate only 3 bins' waste and big trucks can carry up to 6 bins' waste. Another assumption is that, for the conventional WMS, everyday trucks are used for the waste collection whether the bins are full or not. But as the proposed model considers the status of the bins for the link-cost calculation, there is very little chance for the bins' overflowed. Therefore, we have assumed that all bins' collection will be done in 2 days. In other words, instead of collecting all the bin's waste in a single day (conventional system), our model will do the same in 2 days. However, if any bin's status goes to 100% full before the collection, the emergency waste collection will be performed. An alert system might be used when the status of the crosses a threshold value (say for example, 90% full). The smaller truck will be used to collect any immediate waste collection. Nevertheless, the probability of being 100% full of the bin within two days is very low and does not impact the overall performance.

4. RESULTS & DISCUSSION

For the implementation of the algorithm and the model, the values of the distance, bins' status, and the congestion values have been considered randomly (but can be replaced easily for any other cases) in Table 2. By using equation (1), Table 1 and based on values of Table 2, link-cost values have been calculated.

Figure 3 shows the steps behind Algorithm 1. Our starting point is the garage (node 0) and the destination point is the dumping station (node 9). At the first step, node 0 > node 1 > node 6 > node 9 will be chosen based on the Dijkstra's algorithm. After that node 1 and node 6 will be removed from the graph and the link cost will be calculated again. Then again the Dijkstra's algorithm will be used to select the best route. In this way, after 4 rounds, all the nodes will be visited and our algorithm will stop.

For the practical point of view, we have assumed that the operational and fuel costs of the big truck are 0.12 RM per meter and it emits 0.04 gm CO_2 per meter traveling while in the case for the small truck they are 0.05 and 0.02 respectively. The fuel cost and CO_2 emission amount have been taken based on [15] and [16] respectively.

Figure 2, 4-5 show the comparative analysis of the proposed model with the only big truck usage and with the only small truck usage. Figure 2 shows the total distance covered by the trucks. It can be seen that if only small trucks are used, more distance it has to cover. If the proposed model is followed, less distance needs to cover per day collection. The proposed model takes around 26.12% less compared to only a big truck case, while 37.08% less than the only small truck case.

Figure 4 shows that the per day costing of the proposed model is very much less than the only big truck usage and only small truck usage cases. The proposed model costs 59.81% and 41.14% less than those cases respectively.

Figure 5 shows that the proposed model is more eco-friendly than in the other two cases. It emits less CO_2 compared to other cases. The proposed model's CO_2 emission is around 55% and 45.08% lesser than those cases.

Table 2. Values of the parameters and the link cost calculation

from link	to Link	Bin status (%fill levels)	distance (m)	Cong-estion	cost	normalized cost (cost*3)
0 (Garage)	1	70	100	50	1.14	4
0 (Garage)	5	50	200	70	2.96	9
0 (Garage)	9 (dump)	8	100	50	10	30
1	2	50	200	30	2.64	8
1	6	50	80	50	1.36	5
1	7	80	100	30	0.9	3
2	3	80	100	70	1.1	4
2	7	80	50	30	0.52	2
3	4	70	80	50	0.97	3
7	4	80	50	30	0.52	2
5	6	50	100	50	1.6	5
6	8	60	80	50	1.13	4
6	7	80	80	50	0.85	3
8	4	70	80	30	0.85	3
9 (dump)	10	40	100	30	1.8	6
10	11	50	200	50	2.8	9
11	4	60	80	50	1.13	4
10	8	70	70	50	0.88	3
6	9 (dump)	60	100	70	1.46	5

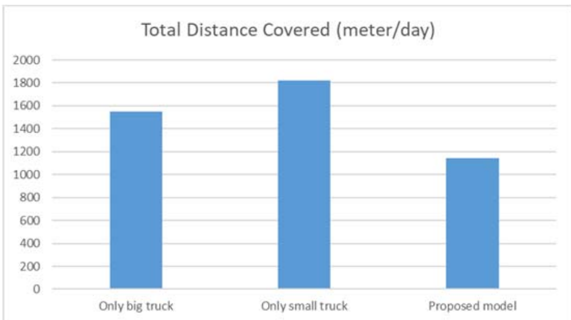


Figure 2. Total distance covered by the trucks in a day.

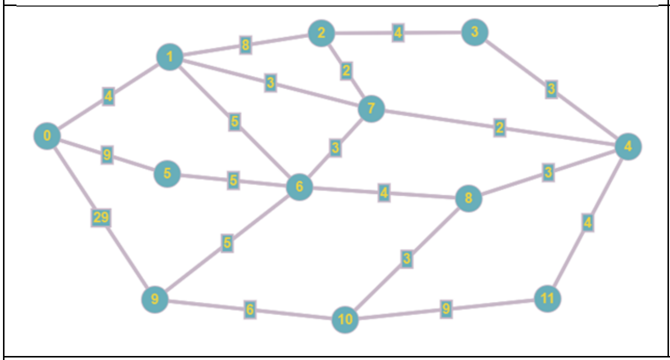
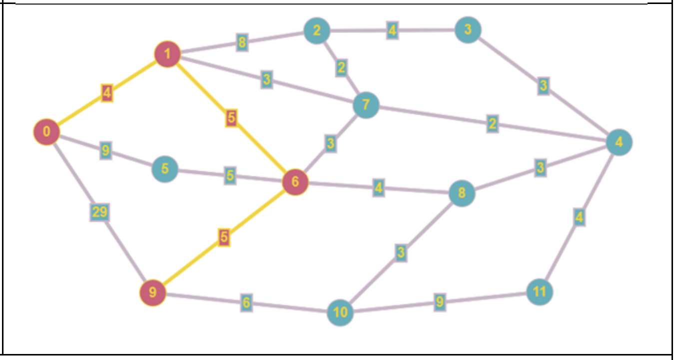
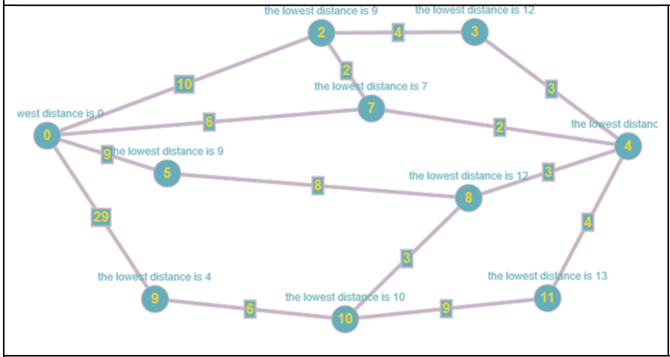
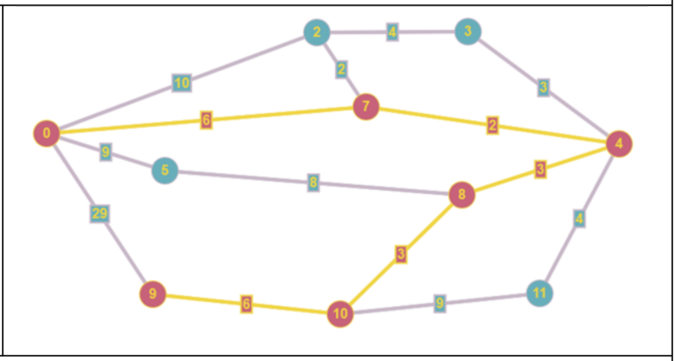
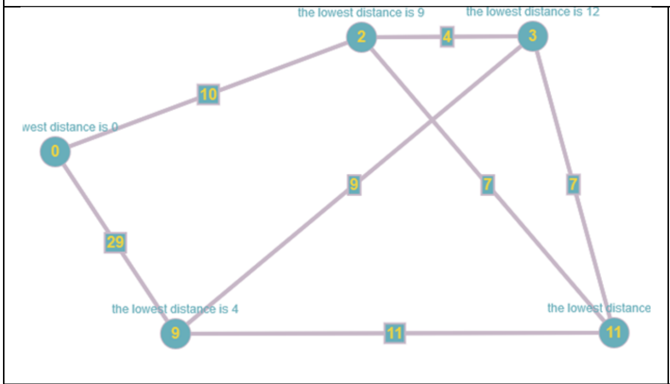
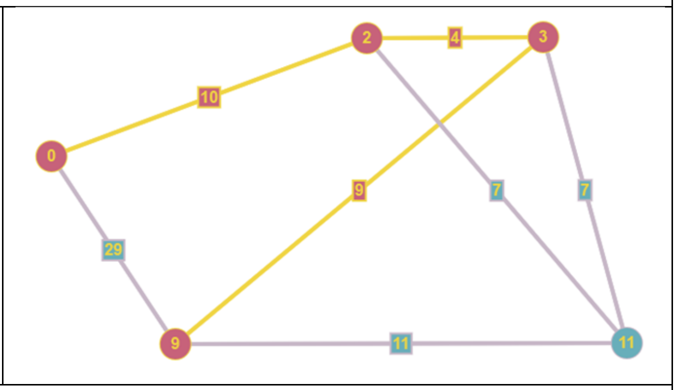
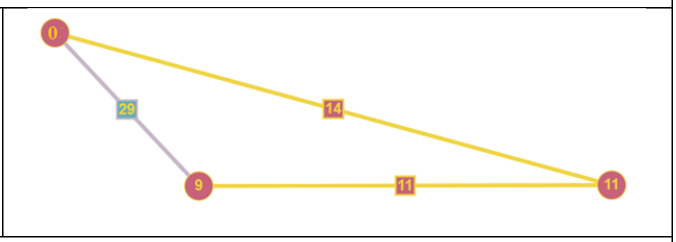
<table>
<tr><td>Initial Bin Placement/Reform after waste collection</td><td>Shortest path selection by using Dijkstra’s Algorithm</td></tr>
<tr><td></td><td></td></tr>
<tr><td colspan="2">Step 1: Garage (0)> Bin 1 > Bin 6>Dumping station (9) = Collections of 2 Bins (by small truck)</td></tr>
<tr><td>
</td><td></td></tr>
<tr><td colspan="2">Step 2: Garage (0)> Bin7 > Bin4> Bin8> Bin10>Dumping station(9) = Collections of 4 Bins (by big truck)</td></tr>
<tr><td>
</td><td></td></tr>
<tr><td colspan="2">Step 3: Garage (0)> Bin2 > Bin3>Dumping station(9) = Collections of 2 Bins (by small truck)</td></tr>
<tr><td></td><td></td></tr>
<tr><td colspan="2">Step 4: Garage (0)> Bin11>Dumping station(9) = Collections of 1 Bins (by small truck)</td></tr>
</table>

Figure 3. Steps of waste collection using Dijkstra’s algorithm.

For the implementation in real-life, the waste collection authority first runs the proposed algorithm with the proposed model. After knowing the best route and the number of bins’ waste to be collected, they will send either small or big garbage truck according to the model and the algorithm. In summary, before deploying the truck for waste collection, they will run the algorithm to know the best route and which truck (small or big) is needed. As the bin’s level is considered for the route selection, there is no chance of the overflow of the bin’s waste. As all the information can be obtained automatically, the whole process would be fully automatic. That means, the route would be automatically determined on a regular basis based on the dynamic changing status of those parameters.

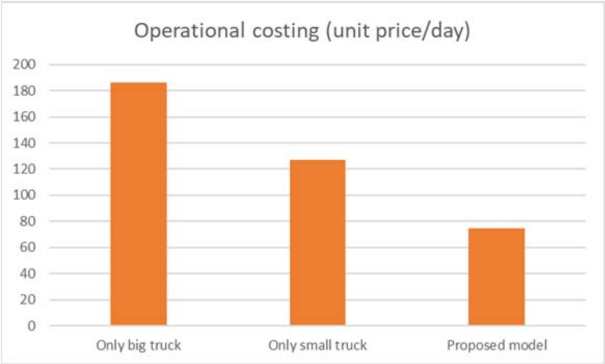


Figure 4. Operational cost comparison.

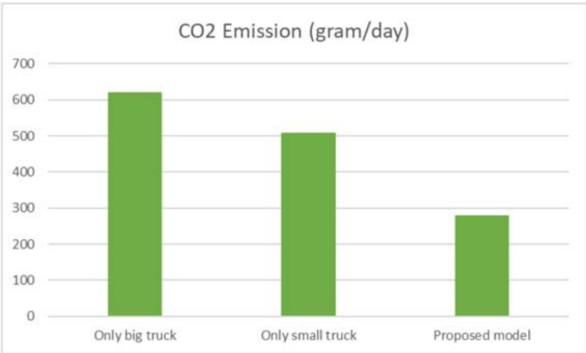


Figure 5. Carbon-dioxide emission comparison.

5. CONCLUSION AND FUTURE WORK

In this paper, a route optimization model and algorithm have been proposed. The algorithm is designed based on Dijkstra's shortest path algorithm. The waste collection problem has been formulated as a single source computer networking problem. Practical values have been considered for the link-costs. The proposed model with the proposed algorithm is found cost-effective, time-efficient and less carbon-dioxide emission.

A prototype of the smart waste management system is under process. After that, this model and algorithm will be verified by using our prototype and the real-life scenarios. Machine learning will be applied to this algorithm for more robust performance.

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